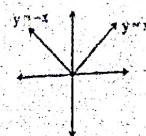


5 Absolute Value Function

$$y = f(x) = |x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$$



1. $|x|^2 = x^2$

2. $\sqrt{x^2} = |x|$

3. $|x| = \max\{-x, x\}$

4. $-|x| = \min\{-x, x\}$

5. $\max(a, b) = \frac{a+b}{2} + \frac{|a-b|}{2}$

6. $\min(a, b) = \frac{a+b}{2} - \frac{|a-b|}{2}$

7. $|x+y| \leq |x| + |y|$

8. $|x+y| = |x| + |y|$ if $xy > 0$

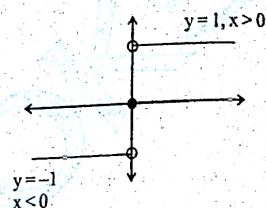
9. $|x-y| = |x| + |y|$ if $xy < 0$

10. $|x| > a$ (is -ve) $x \in \mathbb{R}$

11. $a < |x| < b \Rightarrow b > x > a$ or $a < x < b, x \in [-b, -a] \cup [a, b]$

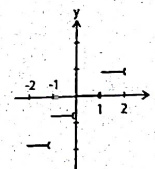
6 Signum Function

$$y = \text{sgn}(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$



7 Greatest Integer Function

$f(x) = [x]$ the integral part of x , which is nearest & smaller integer



1. $[x] \leq x < [x] + 1$

2. $x - 1 < [x] < x$

3. $I \leq x < I + 1 \Rightarrow [x] = I$

4. $[x] - [-x] = \begin{cases} 2x, & x \in \mathbb{I} \\ 2x+1, & x \notin \mathbb{I} \end{cases}$

5. $[x] + [-x] = \begin{cases} 0, & x \in \mathbb{I} \\ -1, & x \notin \mathbb{I} \end{cases}$

6. $[x] \leq n \Leftrightarrow x < n+1, n \in \mathbb{I}$

7. $[x] < n \Leftrightarrow x < n$

8. $[x] = \left[\frac{x}{2} \right] + \left[\frac{x+1}{2} \right]$

9. $\left[\frac{n+1}{2} \right] + \left[\frac{n+2}{4} \right] + \left[\frac{n+4}{8} \right] + \dots = n$

10. $[x] + [y] < [x+y] < [x] + [y] + 1$

11. $[x] + \left[x + \frac{1}{n} \right] + \left[x + \frac{2}{n} \right] + \dots + \left[x + \frac{n-1}{n} \right] = [nx]$

8 Fractional Part Function

$y = \{x\}$ fractional part of x .
 $y = \{x\} = x - [x]$

1. $\{x\} = x, 0 \leq x < 1$

2. $\{x\} = 0, x \in \mathbb{I}$

3. $\{-x\} = 1 - \{x\}, x \in \mathbb{I}$

4. $\{x \neq \text{integer}\} = \{x\}$

9 Odd and Even Function

1. If $f(-x) = -f(x) \forall x \in \mathbb{R}$ then f is an odd function, odd functions are symmetrical in opposite quadrants or about origin.

2. If $f(-x) = f(x)$, then even. It is symmetric about y axis.

- Product of two odd or two even function is an even function.
- Every function can be expressed as the sum of an even and odd function, i.e.,

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Properties

- Product of odd & even function is an odd function.
- Derivative of an odd function is an even function and of an even is odd.

6

Periodic function

$f(x)$ is periodic if $f(x + T) = f(x) \forall x \in \mathbb{R}, T = \text{period}$

Functions	Period
$\sin^2 x, \cos^2 x, \sec^2 x, \csc^2 x$	π (n is even), 2π (n odd/fraction)
$\tan^2 x, \cot^2 x$	π
(trigonometric function)	π
$x - [x]$	1
$f(x) = \text{constant}$	Periodic with no fundamental period.

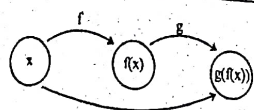
Properties of Periodic functions

If $f(x)$ is periodic with period T , then

- $f(x)$ is periodic with period T
- $f(x + c)$ is periodic with period T .
- $f(x) \pm c$ is periodic with period T .
- $kf(cx + d)$ has period $\frac{T}{|c|}$ period is only affected by coefficient of x .

11

Composition of Function



- $h(x) = g(f(x)) = (g \circ f)(x)$
- $g \circ f \neq f \circ g$
- Composition of two bijection is a bijection.

Properties of Composite Function

f	g	fog
even	even	even
odd	odd	odd
even	odd	even
odd	even	odd

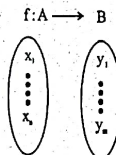
12

Kinds of Mapping

- One-one/injective/Homomorphic:** $f(x) = f(y) \Rightarrow x = y$, then one-one. Graphically, if no line parallel to x-axis meets the graph of function at more than one point.
- Onto/Surjective:** If range = co-domain. Method to show surjectivity: Finding the range of $y = f(x)$ & Showing range of $f = \text{co-domain of } f$
- Many-one mapping:** If two or more elements in domain have same image in co-domain.
- Into Function:** There's an element in B not having a pre image in A under f . [$f: A \rightarrow B$].

13

Number of functions



Total no of functions = m^n

Number of One to one functions = $\begin{cases} {}^m P_n, & m \geq n \\ 0, & m < n \end{cases}$

No. of many one functions = $\begin{cases} m^n - {}^m P_n, & m \geq n \\ m^n, & m < n \end{cases}$

No. of constant function = m

No. of onto function = $\sum_{r=0}^n (-1)^r {}^m C_r (m-r)^n, n \geq m$
 $m^n, n < m$

No. of one-to-one onto functions = $n!$ if $m = n$

14

Inverse of a function

$g: B \rightarrow A, f(x) = y \Leftrightarrow g(y) = x \forall x \in A \text{ and } y \in B$

Then g is Inverse of f

1. Inverse of a bijection is unique.

2. If $f: A \rightarrow B$ is a bijection $g: B \rightarrow A$ is the inverse of f , then $f \circ g = I_B$, where I_A & I_B are Identity function

3. The inverse of a bijection is also a bijection
 $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$

Property -07

$$\begin{aligned} \text{(i)} \sin^{-1} x + \sin^{-1} y &= \sin^{-1} \left\{ x\sqrt{1-y^2} + y\sqrt{1-x^2} \right\}, \text{ if } x^2 + y^2 \leq 1 \\ &\text{ or } x < 0 \text{ and } y > 1, \text{ or } x > 1 \text{ and } y < 0 \\ &= \pi - \sin^{-1} \left\{ x\sqrt{1-y^2} + y\sqrt{1-x^2} \right\}, \text{ if } 0 < x \leq 1, 0 < y \leq 1 \text{ and } x^2 + y^2 > 1 \\ &= -\pi - \sin^{-1} \left\{ x\sqrt{1-y^2} + y\sqrt{1-x^2} \right\}, \text{ if } -1 \leq x < 0, 0 < y \leq 1 \text{ and } x^2 + y^2 > 1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \sin^{-1} x - \sin^{-1} y &= \sin^{-1} \left\{ x\sqrt{1-y^2} - y\sqrt{1-x^2} \right\}, xy > 0, x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 > 1 \\ &= \pi - \sin^{-1} \left\{ x\sqrt{1-y^2} - y\sqrt{1-x^2} \right\}, 0 < x \leq 1, -1 \leq y \leq 0, x^2 + y^2 > 1 \\ &= -\pi - \sin^{-1} \left\{ x\sqrt{1-y^2} - y\sqrt{1-x^2} \right\}, -1 \leq x < 0, 0 < y \leq 1, x^2 + y^2 > 1 \end{aligned}$$

Property -08

$$\begin{aligned} \text{(i)} \cos^{-1} x + \cos^{-1} y &= \cos^{-1} \left\{ xy - \sqrt{1-x^2}\sqrt{1-y^2} \right\}, -1 \leq x, y \leq 1, x+y \geq 0 \\ &= 2\pi - \cos^{-1} \left\{ xy - \sqrt{1-x^2}\sqrt{1-y^2} \right\}, -1 \leq x, y \leq 1, x+y < 0 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \cos^{-1} x - \cos^{-1} y &= \cos^{-1} \left\{ xy + \sqrt{1-x^2}\sqrt{1-y^2} \right\}, -1 \leq x, y \leq 1, x > y \\ &= -\cos^{-1} \left\{ xy + \sqrt{1-x^2}\sqrt{1-y^2} \right\}, -1 \leq x < 0, 0 < y \leq 1, x > y \end{aligned}$$

Property -09

$$\begin{aligned} \text{(i)} 2\sin^{-1} x &= \sin^{-1} \left(2x\sqrt{1-x^2} \right), \frac{-1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ &= \pi - \sin^{-1} \left(2x\sqrt{1-x^2} \right), \frac{1}{\sqrt{2}} \leq x \leq 1 \\ &= -\pi - \sin^{-1} \left(2x\sqrt{1-x^2} \right), -1 \leq x \leq \frac{-1}{\sqrt{2}} \\ \text{(ii)} 3\sin^{-1} x &= \sin^{-1} (3x - 4x^3), -1/2 \leq x \leq 1/2 \\ &= \pi - \sin^{-1} (3x - 4x^3), 1/2 < x \leq 1 \\ &= -\pi - \sin^{-1} (3x - 4x^3), -1 \leq x < -1/2 \end{aligned}$$

Property -10

$$\begin{aligned} \text{(i)} 2\cos^{-1} x &= \cos^{-1} (2x^2 - 1), 0 \leq x \leq 1 = 2\pi - \cos^{-1} (2x^2 - 1), -1 \leq x \leq 0 \\ \text{(ii)} 3\cos^{-1} x &= \cos^{-1} (4x^3 - 3x), 1/2 \leq x \leq 1 \\ &= 2\pi - \cos^{-1} (4x^3 - 3x), -1/2 \leq x \leq 1/2 \\ &= 2\pi + \cos^{-1} (4x^3 - 3x), -1 \leq x \leq -1/2 \end{aligned}$$

Property -11

$$\begin{aligned} \text{(i)} 2\tan^{-1} x &= \tan^{-1} \left(\frac{2x}{1-x^2} \right), -1 < x < 1 \\ &= \pi + \tan^{-1} \left(\frac{2x}{1-x^2} \right), x > 1 \\ &= -\pi + \tan^{-1} \left(\frac{2x}{1-x^2} \right), x < -1 \\ \text{(ii)} 2\tan^{-1} x &= \sin^{-1} \left(\frac{2x}{1+x^2} \right), -1 \leq x \leq 1 \\ &= \pi - \sin^{-1} \left(\frac{2x}{1+x^2} \right), x > 1 \\ &= -\pi - \sin^{-1} \left(\frac{2x}{1+x^2} \right), x < -1 \\ \text{(iv)} 3\tan^{-1} x &= \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right), \frac{-1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ &= \pi + \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right), x > \frac{1}{\sqrt{3}} \\ &= -\pi + \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right), x < \frac{-1}{\sqrt{3}} \\ \text{(iii)} 2\tan^{-1} x &= \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), 0 \leq x < \infty \\ &= -\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), -\infty < x \leq 0 \end{aligned}$$

Property -12

$$\begin{aligned} \text{(i)} \sin^{-1} x &= \cos^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{x}{\sqrt{1-x^2}} = \cot^{-1} \frac{\sqrt{1-x^2}}{x} = \sec^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) = \operatorname{cosec}^{-1} \left(\frac{1}{x} \right), x > 0 \\ \text{(ii)} \cos^{-1} x &= \sin^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{\sqrt{1-x^2}}{x} = \cot^{-1} \frac{x}{\sqrt{1-x^2}} = \sec^{-1} \left(\frac{1}{x} \right) = \operatorname{cosec}^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right), x > 0 \\ \text{(iii)} \tan^{-1} x &= \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) = \cot^{-1} \left(\frac{1}{x} \right) = \sec^{-1} \left(\sqrt{1+x^2} \right) = \operatorname{cosec}^{-1} \left(\frac{\sqrt{1+x^2}}{x} \right) \end{aligned}$$

Property -13

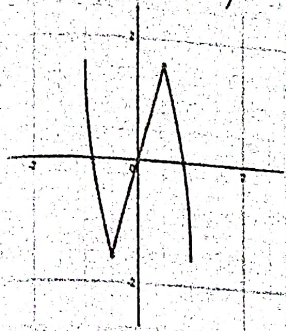
If $x_1, x_2, \dots, x_n \in \mathbb{R}$ then $\tan^{-1} x_1 + \tan^{-1} x_2 + \dots + \tan^{-1} x_n =$

$$\tan^{-1} \left(\frac{s_1 - s_2 + s_3 - s_4 + \dots}{1 - s_2 + s_4 - s_6 + \dots} \right)$$

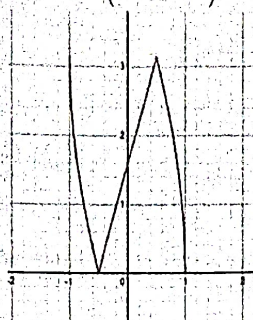
where, s_k = sum of products of $x_1, x_2, \dots, x_n$ taken k at a time.

Some Important graphs

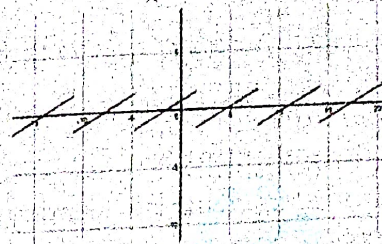
$$\sin^{-1}(3x - 4x^3)$$



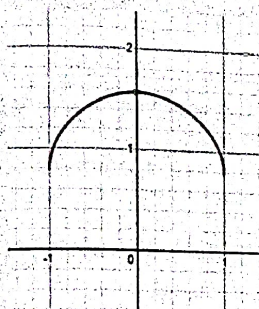
$$\cos^{-1}(4x^3 - 3x)$$



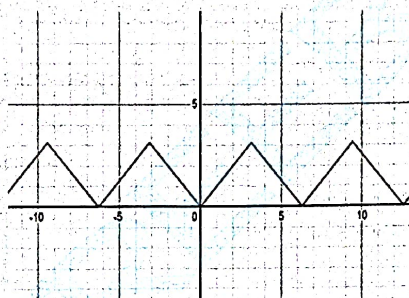
$$\tan^{-1}(\sec x + \tan x)$$



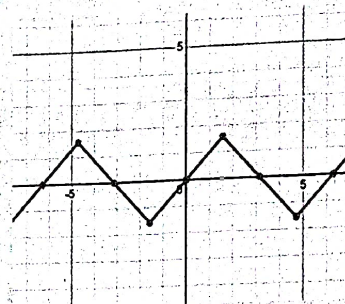
$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \frac{\pi}{4} + \frac{1}{2}\cos^{-1}x^2$$



$$\cos^{-1}(\cos x)$$



$$\sin^{-1}(\sin x)$$



MATRICES

01. MATRIX

A matrix is an ordered rectangular array of numbers or functions. The numbers or functions are called the elements of the matrix

02. ORDER OF A MATRIX

A matrix having m rows and n columns is called a matrix of order $m \times n$ or simply $m \times n$ matrix.

$$\text{or } A = [a_{ij}]_{m \times n}, 1 \leq i \leq m, 1 \leq j \leq n, i, j \in \mathbb{N}$$

a_{ij} is an element lying in the i^{th} row & j^{th} column. The number of elements in $m \times n$ matrix will be mn .

03. TYPE OF MATRIX

(i) **Column Matrix:** A matrix is said to be a column matrix if it has only one column, i.e., $A = [a_{ij}]_{m \times 1}$ is a column matrix of order $m \times 1$.

(ii) **Row Matrix:** Row matrix has only one row, i.e., $B = [b_{ij}]_{1 \times n}$ is a row matrix of order $1 \times n$.

(iii) **Square Matrix:** Square matrix has equal number of rows and columns, i.e., $A = [a_{ij}]_{m \times m}$ is a square matrix of order m .

(iv) **Diagonal Matrix:** A square matrix is said to be diagonal matrix if all of its non-diagonal elements are zero, i.e., $B = [b_{ij}]_{m \times m}$ is said to be a diagonal matrix if $b_{ij} = 0$, where $i \neq j$.

(v) **Scalar Matrix:** It is a diagonal matrix with all its diagonal elements are equal, i.e., $B = [b_{ij}]_{m \times m}$ is a scalar matrix if $b_{ij} = 0$, where $i \neq j$, $b_{ii} = k$, when $i = j$ & $k = \text{constant}$.

(vi) **Identity Matrix:** It is a diagonal matrix having all its diagonal elements equal to 1, i.e., $A = [a_{ij}]_{m \times m}$ is an identity matrix if

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

we denote identity matrix by I_n when order is n .

(vii) **Zero Matrix:** A matrix is said to be zero or null matrix if all its elements are zero. It is denoted by O .

04. EQUALITY OF MATRICES

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if

- they are of the same order
- each element of A is equal to the corresponding element of B , i.e., $a_{ij} = b_{ij}$ for all i & j

05. TRACE OF A MATRIX

The sum of diagonal element of a square matrix A is called the trace of matrix A , which is denoted by $\text{tr } A$

$$\text{tr } A = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

Properties of Trace of a Matrix

Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ and λ be a scalar.

$$(i) \text{tr}(\lambda A) = \lambda \text{tr}(A) \quad (ii) \text{tr}(A - B) = \text{tr}(A) - \text{tr}(B)$$

$$(iii) \text{tr}(AB) = \text{tr}(BA) \quad (iv) \text{tr}(I_n) = n$$

$$(vi) \text{tr}(AB) \neq \text{tr } A \cdot \text{tr } B \quad (v) \text{tr}(O) = 0$$

06. ADDITION OF MATRICES

Properties of matrix Addition

(i) **Commutative Law:** $A + B = B + A$ (ii) **Associative Law:** $(A + B) + C = A + (B + C)$

(iii) **Existence of Additive Identity:** Let $A = [a_{ij}]_{m \times n}$ & O = zero matrix of order $m \times n$, then $A + O = O + A = A$. Here O is the additive identity for matrix addition.

(iv) **Existence of Additive Inverse:** Let $A = [a_{ij}]_{m \times n}$ be any matrix then we have another matrix as Let $-A = [-a_{ij}]_{m \times n}$ such that $A + (-A) = (-A) + A = O$. Here $-A$ is the additive inverse of A or negative of A .

07. MULTIPLICATION OF A MATRIX BY A SCALAR

Let $A = [a_{ij}]_{m \times n}$ be a matrix & k, t be a number.

$$\text{Then, } kA = Ak = [ka_{ij}]_{m \times n}$$

Properties

$$(i) k(A + B) = kA + kB$$

$$(ii) (k + t)A = kA + tA$$

08. MULTIPLICATION OF MATRICES

If A & B are any two matrices, then their product will be defined only when the number of columns in A is equal to the number of rows in B .

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$ then their product $AB = C = [c_{ij}]$ is a matrix of order, $m \times p$ where

$$(ij)^{\text{th}} \text{ element of } AB = C_{ij} = \sum_{r=1}^n a_{ir} b_{rj}$$

9. PROPERTIES OF MATRIX MULTIPLICATION

- (i) **Associative Law for Multiplication:** If A, B & C are three matrices of order $n \times n, n \times p$ & $p \times q$ respectively, then $(AB)C = A(BC)$
- (ii) **Distributive Law:** For three matrices A, B & C (a) $A(B + C) = AB + AC$
(b) $(A + B)C = AC + BC$ whenever both sides of equality are defined.
- (iii) Matrix Multiplication is not commutative in general, i.e. $AB \neq BA$ (in general).
- (iv) **Existence of Multiplicative Identity:** For every square matrix, there exists an identity matrix I of same order such that $IA = AI = A$

10. PROPERTIES OF TRANSPOSE OF THE MATRICES

For any matrices A & B of suitable orders, we have:

- (i) $(A^T)^T = A$
(ii) $(kA)^T = k(A^T)$ (where k is constant)
(iii) $(A \pm B)^T = A^T \pm B^T$
(iv) $(AB)^T = B^T A^T$
(v) $(A_1 A_2 A_3 \dots A_{n-1} A_n)^T = A_n^T A_{n-1}^T \dots A_2^T A_1^T$
(vi) $I^T = I$

11. MATRIX POLYNOMIAL

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_{n-1} x + a_n$ be a polynomial and let A be a square matrix of order n . Then $f(A) = a_n A^n + a_{n-1} A^{n-1} + a_{n-2} A^{n-2} + \dots + a_{n-1} A + a_n I_n$ is called a matrix polynomial.

13. INVERTIBLE MATRIX AND INVERSE MATRIX

Properties of Invertible Matrices

- (i) **Uniqueness of Inverse:** Inverse of a square matrix, if it exists, is unique.
- (i) $(A^{-1})^{-1} = A$ (ii) $(A^T)^{-1} = (A^{-1})^T$
(iii) $(AB)^{-1} = B^{-1} A^{-1}$ (iv) $(A^T)^{-1} = (A^{-1})^T$

15. IDEMPOTENT MATRIX

A square matrix A is called an idempotent matrix if $A^2 = A$.

Example: $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$ is an idempotent matrix, because

$$A^2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = A$$

Also, $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ are idempotent matrices because $A^2 = A$ and $B^2 = B$.

17. PERIODIC MATRIX

A matrix A will be called a periodic matrix if $A^{k+1} = A$ where k is a positive integer. If, however k is the least positive integer for which $A^{k+1} = A$, then k is said to be the period of A .

12. SYMMETRIC & SKEW SYMMETRIC MATRICES

Symmetric Matrix

A square matrix $A = [a_{ij}]$ is called a symmetric matrix, if $a_{ij} = a_{ji}$ for all i, j or $A^T = A$

Skew Symmetric Matrix

A square matrix $A = [a_{ij}]$ is called a skew-symmetric matrix, if $a_{ij} = -a_{ji}$ for all i, j or $A^T = -A$

Properties of Symmetric & Skew Symmetric Matrices

- (i) For any square matrix A with real number entries $(A + A^T)$ is a symmetric matrix & $(A - A^T)$ is a skew symmetric matrix.
- (ii) Any square matrix A can be expressed as the sum of a symmetric & a skew symmetric matrix as $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

14. ORTHOGONAL MATRIX

A square matrix A is called orthogonal if $AA^T = I = A^T A$, i.e., if $A^{-1} = A^T$.

Example: $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = A^T$. In fact every unit matrix is orthogonal

16. INVOLUTORY MATRIX

A square matrix A is called an involutory matrix if $A^2 = I$ or $A^{-1} = A$

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is an involutory matrix because

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \text{ In fact every unit matrix is involutory}$$

18. NILPOTENT MATRIX

A square matrix A is called a nilpotent matrix if there exists $p \in \mathbb{N}$ such that $A^p = 0$.

Example: $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ is a nilpotent matrix because $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$ (Here $p = 2$)

19. DIFFERENTIATION OF A MATRIX

If $A = \begin{bmatrix} f(x) & g(x) \\ h(x) & l(x) \end{bmatrix}$, then $\frac{dA}{dx} = \begin{bmatrix} f'(x) & g'(x) \\ h'(x) & l'(x) \end{bmatrix}$ is a differentiation of matrix A .

DETERMINANT

1.

DETERMINANT OF A SQUARE MATRIX OF ORDER TWO AND THREE

Expansion of two order: $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - b_1 a_2$

Expansion of three order: $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$

2.

SARRUS RULE

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta = P - N$$

3.

PROPERTIES OF DETERMINANTS

- The value of a determinant remains unchanged if its rows and columns are interchanged.
- If any two rows (or columns) of a determinant are interchanged, then sign of determinant changes.
- If any two rows (or columns) of a determinant are identical, then the value of determinant is zero.
- If each element of a row (or a column) of a determinant is multiplied by a constant k , then its value gets multiplied by k .
- If some or all the elements of a row or column of a determinant are expressed as a sum of two (or more) terms, then the determinant can be expressed as a sum of two (or more) determinants.
- If the equimultiples of corresponding elements of other row (or column) are added to each element of any row or column of a determinant, then the value of the determinant remains the same.
- $|A'| = |A|$, where A' is transpose of A .
- If $A = [a_{ij}]_{3 \times 3}$, then $|kA| = k^3 |A|$.
- The determinant of the product of matrices is equal to product of their respective determinants, i.e., $|AB| = |A||B|$, where A & B are square matrices of same order.

$$(x) \begin{vmatrix} a_1 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & 0 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ a_2 & b_2 & 0 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ 0 & b_2 & 0 \\ 0 & 0 & c_3 \end{vmatrix} = a_1 b_2 c_3$$

4.

USE OF DETERMINANTS IN CO-ORDINATE GEOMETRY

(i) Area of triangle, whose vertices are $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

(ii) If $ax + by + c = 0$ are the sides of a triangle, then the area =

$$\frac{1}{2c_1 c_2 c_3} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 \quad c_1, c_2, c_3 \text{ are cofactors of } c_1, c_2, c_3$$

(iii) Equation of a straight line passing through two points

$$(x_1, y_1) \text{ \& } (x_2, y_2) \text{ is } \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

(iv) If three lines $ax + by + c = 0$ are concurrent, then $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.

(v) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight

$$\text{line then } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

(vi) The equation of circle through three non-collinear points (x_1, y_1) is

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0$$

5.

MINOR AND COFACTOR OF AN ELEMENT OF A DETERMINANT

Minor: The determinant that is left by cancelling the row and column intersecting at a particular element of a determinant is called the minor of that element of the determinant. Minor of an element a_{ij} of a determinant is denoted by M_{ij} .

Cofactor: The cofactor of an element a_{ij} of a determinant is denoted by A_{ij} (or C_{ij}) and is equal to $(-1)^{i+j} M_{ij}$.

6.

ADJOINT OF A MATRIX

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then $\text{adj } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$, where A_{ij} is the cofactor of a_{ij}

If A be any given square matrix of order n , then $A(\text{adj } A) = (\text{adj } A)A = |A|I$ where I is the identity matrix of order n .

7.

PROPERTIES OF ADJOINT OF A MATRIX

If A be any given square matrix of order n , we can define the following:

- $A(\text{adj } A) = (\text{adj } A)A = |A|I$, where I is the identity matrix of order n
- For a zero matrix O , $\text{adj}(O) = O$
- For an identity matrix I , $\text{adj}(I) = I$
- For any scalar k , $\text{adj}(kA) = k^{n-1} \text{adj}(A)$
- $\text{adj}(A') = (\text{adj } A)'$
- $\det(\text{adj } A)$, i.e. $|\text{adj } A| = (\det A)^{n-1}$
- If A is an invertible matrix and A^{-1} be its inverse, then: $\text{adj } A = (\det A)^{n-1} A^{-1}$ is invertible with inverse $(\det A)^{-1} \text{adj}(A^{-1}) = (\text{adj } A)^{-1}$
- Suppose A and B are two matrices of order n , then $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$
- For any non-negative integer p , $\text{adj}(A^p) = (\text{adj } A)^p$
- If A is invertible, then the above formula also holds for negative k .

8.

SINGULAR AND NON-SINGULAR MATRICES

A square matrix A is said to be singular if $|A| = 0$, otherwise it is called non-singular matrix.

If A & B are non-singular matrix of same order, then AB & BA are also non-singular matrices of same order.

9.

INVERSE OF A MATRIX

If A and B are two matrices such that $AB = I = BA$, then B is called the inverse of A and it is denoted by A^{-1}

Also, $A^{-1} = \frac{\text{adj } A}{|A|}$, if $|A| \neq 0$

10.

PROPERTIES OF INVERSE

If A and B are the non-singular matrices, then the inverse matrix should have the following properties:

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
- $(A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$
- $(kA)^{-1} = (1/k)A^{-1}$
- $AB = I_n$ where A and B are inverse of each other.

11.

PRODUCT OF TWO DETERMINANTS

$$\text{Let } \Delta_1 = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \text{ and } \Delta_2 = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}$$

Then product of Δ_1 and Δ_2 is defined as

$$\Delta_1 \Delta_2 = \begin{vmatrix} a_1x_1 + a_2x_2 + a_3x_3 & a_1y_1 + a_2y_2 + a_3y_3 & a_1z_1 + a_2z_2 + a_3z_3 \\ b_1x_1 + b_2x_2 + b_3x_3 & b_1y_1 + b_2y_2 + b_3y_3 & b_1z_1 + b_2z_2 + b_3z_3 \\ c_1x_1 + c_2x_2 + c_3x_3 & c_1y_1 + c_2y_2 + c_3y_3 & c_1z_1 + c_2z_2 + c_3z_3 \end{vmatrix}$$

12.

DIFFERENTIATION OF A DETERMINANT

$$\text{Let } \Delta(x) = \begin{vmatrix} f_1(x) & g_1(x) & h_1(x) \\ f_2(x) & g_2(x) & h_2(x) \\ f_3(x) & g_3(x) & h_3(x) \end{vmatrix}$$

then $\Delta'(x)$

$$= \begin{vmatrix} f_1'(x) & g_1'(x) & h_1'(x) \\ f_2(x) & g_2(x) & h_2(x) \\ f_3(x) & g_3(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) & h_1'(x) \\ f_2'(x) & g_2'(x) & h_2'(x) \\ f_3(x) & g_3(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1(x) & h_1'(x) \\ f_2(x) & g_2(x) & h_2'(x) \\ f_3'(x) & g_3'(x) & h_3'(x) \end{vmatrix}$$

13.

INTEGRATION OF DETERMINATION

$$\text{Let } \Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ p & q & r \\ l & m & n \end{vmatrix}, \text{ where } p, q, r, l, m \text{ and } n \text{ are constants.}$$

$$\text{Then } \int_a^b \Delta(x) dx = \begin{vmatrix} \int_a^b f(x) dx & \int_a^b g(x) dx & \int_a^b h(x) dx \\ p & q & r \\ l & m & n \end{vmatrix}$$

14.

USE OF SUMMATION

$$\text{If } f(r) = \begin{vmatrix} r & r^2 & r^3 \\ p & q & t \\ 1 & 2 & 3 \end{vmatrix} \text{ where } p, q, t \text{ are constants, then } \sum_{r=1}^n f(r) = \begin{vmatrix} \sum_{r=1}^n r & \sum_{r=1}^n r^2 & \sum_{r=1}^n r^3 \\ p & q & t \\ 1 & 2 & 3 \end{vmatrix}$$

16.

Check value of Δ

$$\Delta \neq 0$$

Consistent system and has unique solution

$$x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

$$\Delta = 0$$

Check the values of Δ_1, Δ_2 and Δ_3

At least one of Δ_1, Δ_2 and Δ_3 is not zero

Inconsistent system

$$\Delta_1 = \Delta_2 = \Delta_3 = 0$$

Put $z = t$ and solve any two equations to get the values of x and y in terms of t .

15.

$$\begin{aligned} ax + by + c_1 &= 0 \\ ax + by + c_2 &= 0 \end{aligned}$$

Consistent system of equation has solution

Inconsistent system of equation has no solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

(Equation represents intersecting lines)

Infinito solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

(Equation represents coincident lines)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

(Equation represents parallel disjoint lines)

17.

1. Symmetric determinant

The elements situated at equal distance from the diagonal are equal both in magnitude and sign.

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2$$

2. Skew symmetric determinant

All the diagonal elements are zero and the elements situated at equal distance from the diagonal are equal in magnitude but opposite in sign. The value of a skew symmetric determinant of odd order is zero.

$$\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$$

3. Circulant determinant:

The elements of the rows (or columns) are in cyclic arrangement

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$$

$$4. \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$5. \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$6. \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

LIMITS

1.

Right hand and Left hand Limit

To evaluate $\lim_{x \rightarrow a^+} f(x)$

1. Put $x = a + h$ in $f(x)$ to get $\lim_{h \rightarrow 0} f(a+h)$

2. Take the limit as $h \rightarrow 0$.

To evaluate $\lim_{x \rightarrow a^-} f(x)$

1. Put $x = a - h$ in $f(x)$ to get $\lim_{h \rightarrow 0} f(a-h)$

2. Take the limit as $h \rightarrow 0$.

2.

Some Useful Limits

(1) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, where $n \in \mathbb{Q}$, the set of rational numbers.

(2)

$$(i) \lim_{n \rightarrow \infty} a^n = \begin{cases} \infty, & \text{if } a > 1 \\ 1, & \text{if } a = 1 \\ 0, & \text{if } -1 < a < 1 \\ \text{does not exist,} & \text{if } a \leq -1 \end{cases}$$

$$(ii) \lim_{x \rightarrow \infty} \frac{a_0 x^p + a_1 x^{p-1} + \dots + a_{p-1} x + a_p}{b_0 x^q + b_1 x^{q-1} + \dots + b_{q-1} x + b_q}$$

$$= \begin{cases} \frac{a_0}{b_0}, & \text{if } p = q \\ 0, & \text{if } p < q \\ \infty, \frac{a_0}{b_0} > 0 & \text{if } p > q \end{cases}$$

(3) Trigonometric Limits

$$(i) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$(ii) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$(iii) \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$$

$$(iv) \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$$

$$(v) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\pi}{180}$$

Some Useful Expansions

$$(i) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \text{ to } \infty$$

$$(ii) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \text{ to } \infty$$

$$(iii) \tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots \text{ to } \infty$$

$$(iv) \sin^{-1} x = x + \frac{1^2 x^3}{3!} + \frac{1^2 \cdot 3^2 x^5}{5!} + \frac{1^2 \cdot 3^2 \cdot 5^2 x^7}{7!} + \dots \text{ to } \infty$$

$$(v) (\sin^{-1} x)^2 = \frac{2}{2!} x^2 + \frac{2 \cdot 2^2}{4!} x^4 + \frac{2 \cdot 2^2 \cdot 4^2}{6!} x^6 + \dots \text{ to } \infty$$

$$(vi) \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \text{ to } \infty$$

(4) Exponential and Logarithmic Limits

$$(i) \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$(ii) \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a, a > 0$$

$$(iii) \lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \log_e \left(\frac{a}{b} \right); a, b > 0$$

$$(iv) \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$$

$$(v) \lim_{n \rightarrow 0} \left(1 + \frac{1}{n} \right)^n = e$$

$$(vi) \lim_{h \rightarrow 0} (1+ah)^{1/h} = e^a$$

$$(vii) \lim_{x \rightarrow \infty} \frac{\log x}{x^m} = 0, (m > 0)$$

$$(viii) \lim_{x \rightarrow 0} \frac{\log_a (1+x)}{x} = \log_a e, (a > 0, a \neq 1)$$

$$(ix) \lim_{x \rightarrow 0} \left(1 + \frac{a}{x} \right)^x = e^a$$

$$(x) \lim_{x \rightarrow 0} \left(1 + \frac{1}{f(x)} \right)^{f(x)} = e, \text{ where } f(x) \rightarrow \infty \text{ as } x \rightarrow 0.$$

$$(xi) \lim_{x \rightarrow 0} (1+f(x))^{1/f(x)} = e, \text{ where } f(x) \rightarrow 0 \text{ as } x \rightarrow 0$$

Some Useful Expansions

$$(i) e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \text{ to } \infty$$

$$(ii) e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \text{ to } \infty$$

$$(iii) \log_e (1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \text{ to } \infty, -1 < x \leq 1$$

$$(iv) \log_e (1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots \text{ to } \infty, -1 \leq x < 1$$

$$(v) a^x = e^{x \log a} = 1 + x \log a + \frac{(x \log a)^2}{2!} + \dots \text{ to } \infty.$$

$$(vi) (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots \text{ to } \infty, -1 < x < 1,$$

Note

$$(i) \text{ If } \lim_{x \rightarrow a} f(x) = A > 0 \text{ and } \lim_{x \rightarrow a} g(x) = B, \text{ then } \lim_{x \rightarrow a} [f(x)]^{g(x)} = A^B.$$

$$(ii) \text{ If } \lim_{x \rightarrow 0} f(x) = 1 \text{ and } \lim_{x \rightarrow 0} g(x) = \infty, \text{ then}$$

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x)[f(x)-1]}.$$

3.

EVALUATION OF LIMITS USING L'HOSPITAL'S RULE

$$(i) \left(\frac{0}{0}\right) \text{ form: If } \lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \text{ provided the limit on the R.H.S. exists.}$$

$$(ii) \left(\frac{\infty}{\infty}\right) \text{ form: If } \lim_{x \rightarrow a} f(x) = \infty \text{ and } \lim_{x \rightarrow a} g(x) = \infty, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \text{ provided the limit on the R.H.S. exists.}$$

Note: That sometimes we have to repeat the process if the form is $\frac{0}{0}$ or $\frac{\infty}{\infty}$ again.